

Lannik Psi rectangular RX prototype: staggered or not?

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Sources: `studies/2026-09-02_lannik-psi/` (`RX_LAYOUT.md` , `MIMO.md` , `NOTES.md` ,
`ANTENNA_REQUIREMENTS.md`)

What has to be decided

- Two antenna prototypes planned; the square-subarray 2×4 layout is likely fixed
- The rectangular prototype ($2.42\lambda \times 4.83\lambda$ subarrays, 4×2 channels) needs its RX phase-centre layout fixed before ordering:
 - unstaggered URA, or
 - alternating columns displaced vertically ("stagger"), and by how much
- The rectangle keeps 3 dB more RX gain than the square but has closer vertical aliases
- This is a recommendation under uncertainty from an idealized model: fixed 1 m^2 target, nominal patterns, no tracker, no measured latency

The ambiguity problem

- Eight channel phase centres form a sparse array; the steering vector repeats every 0.414 in u ($\pm 11.9^\circ$) and 0.207 in v ($\pm 5.9^\circ$)
- The vertical alias pair sits **inside** the TX 3 dB beam ($\pm 6.25^\circ$): it affects targets at long range (1 m² detectable beyond 900 m at the edge)
- The horizontal pair sits outside the beam: for a 1 m² target it is a short-range problem (about 550 m at the 10 dB SNR reference); low-Pd tracking and stronger targets extend that region
- A coherent-TX measurement cannot tell a target from its alias by the steering vector at all
- Gain plausibility (RCS ceiling +10 dBsm) protects less than expected: only about $\pm 2^\circ$ around the horizontal plane, because the *skirt* of the alias lobe becomes admissible before the exact alias does

What MIMO actually does on this antenna

- TX radiator pitch equals RX pitch, so an 8×8 TX quadrant is exactly one RX subarray height: **quadrant geometry gives no discrimination**
- The supplied excitation has a quadratic phase (about 180° at edge centres): a defocus that widens the beam from 6.9° to 12.6° (5.3 dB directivity cost) and squints each quadrant beam 6° toward the opposite corner
- That squint is the entire source of the MIMO alias discrimination

TX quadrant excitation	Vertical edges	Horizontal edges
Nominal (amplitude and phase)	-29.3 dB	-3.1 dB
Amplitude taper only	-4.7 dB	-0.04 dB
Uniform quadrants	0.0 dB	-0.6 dB

MIMO here is a four-squinted-beam comparison, and a property of the TX excitation.

MIMO robustness: fine if the defocus is built, fragile to a different design

- Removing the phase entirely (focused, 6.9° beam): vertical margin 6.1 dB, horizontal 0.0 dB
- Scaling the amplitude taper between the 0.5 and 2 power keeps the vertical-edge correlation below -18 dB
- An uncalibrated 30° upper/lower quadrant phase error still leaves 13 dB vertical margin
- Sensitivity to the *amount* of defocus: a review found a phase-wrapping bug in that sweep. A corrected preliminary check gives a vertical margin above 15 dB for 0.6 to 1.4 times the prescribed phase, but the horizontal margin may drop to about 2 dB at 0.6 times. **To be re-run; treat as unverified**
- The horizontal edge is the MIMO-limited case in every variant
- A spec of gain, beamwidth and sidelobes alone can be met by amplitude-only shaping or a smaller focused aperture, which would leave MIMO with nothing

What a stagger does

- Alternating columns differ by phase $2\pi \cdot s/H$ at the old vertical alias; the two column groups give $\rho = \cos^2(\pi \cdot s/H)$
- Our sketch used one radiator pitch ($H/8$), chosen without analysis

Offset s	Coherent-mode separation $1 - \rho$	Horizontal aliases	RX height
0 (URA)	0.00	exact	37.6 mm
$H/8$, one pitch	0.15	exact	40.0 mm
$H/4$, two pitches	0.50	exact	42.3 mm
$H/2$	1.00	exact	47.0 mm

- No change to subarray area, gain, beamwidth or sensitivity envelope
- Horizontal aliases stay exact whatever the offset

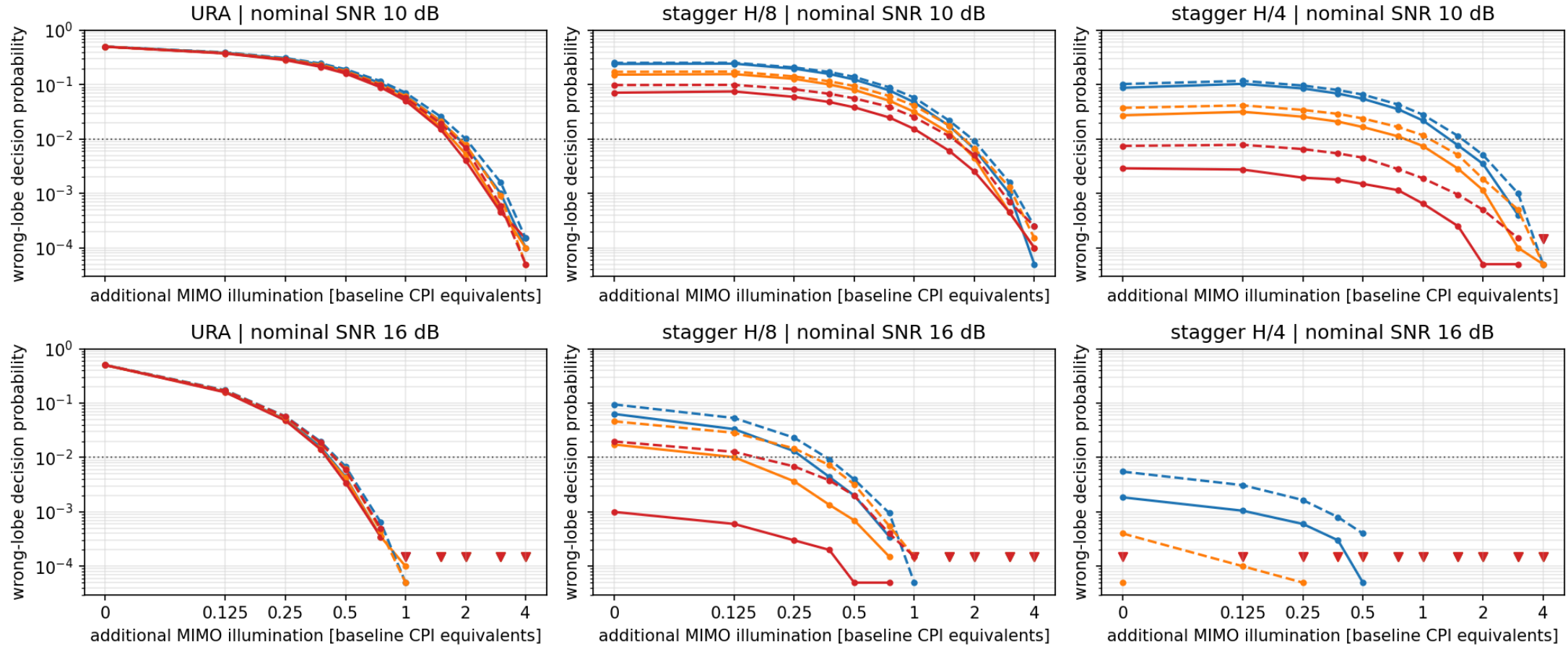
Resolution events: coherent frames only, vertical edge, 1 m² target

Wrong-lobe probability (fixed RCS, independent frame noise, forced choice):

SNR (range)	Frames	URA	H/8	H/4
10 dB (893 m)	4	0.50	0.071	0.0029
10 dB, 10° phase error	4	0.50	0.099	0.0075
16 dB (632 m)	1	0.50	0.063	0.0019
16 dB, 10° phase error	4	0.50	0.020	0.0000

- Four frames are 200 ms at 20 Hz with no mode switch
- H/4 meets a 1% objective with **no MIMO burst** in every tested vertical case; H/8 still needs a burst at 893 m
- Horizontal-edge and corner events are identical for all three layouts (MIMO needed: 1.5 CPI at 10 dB, 0.375 at 16 dB)
- Promising evidence, **not a resolution-time guarantee**: fixed RCS, forced choice, no motion, and the candidate set does not yet include the diagonal near-alias family that a stagger creates

Vertical-edge resolution event: K coherent frames + on-demand MIMO
URA versus height/8 and height/4 alternating column stagger



Fixed 1 m² target in a known gate; 20,000 trials; independent frame noise; unknown amplitude per update; ▼: zero errors at 95% upper bound
Ideal energy scaling, not latency; motion or detection threshold: height/4 is a study construct, not a supplier proposal

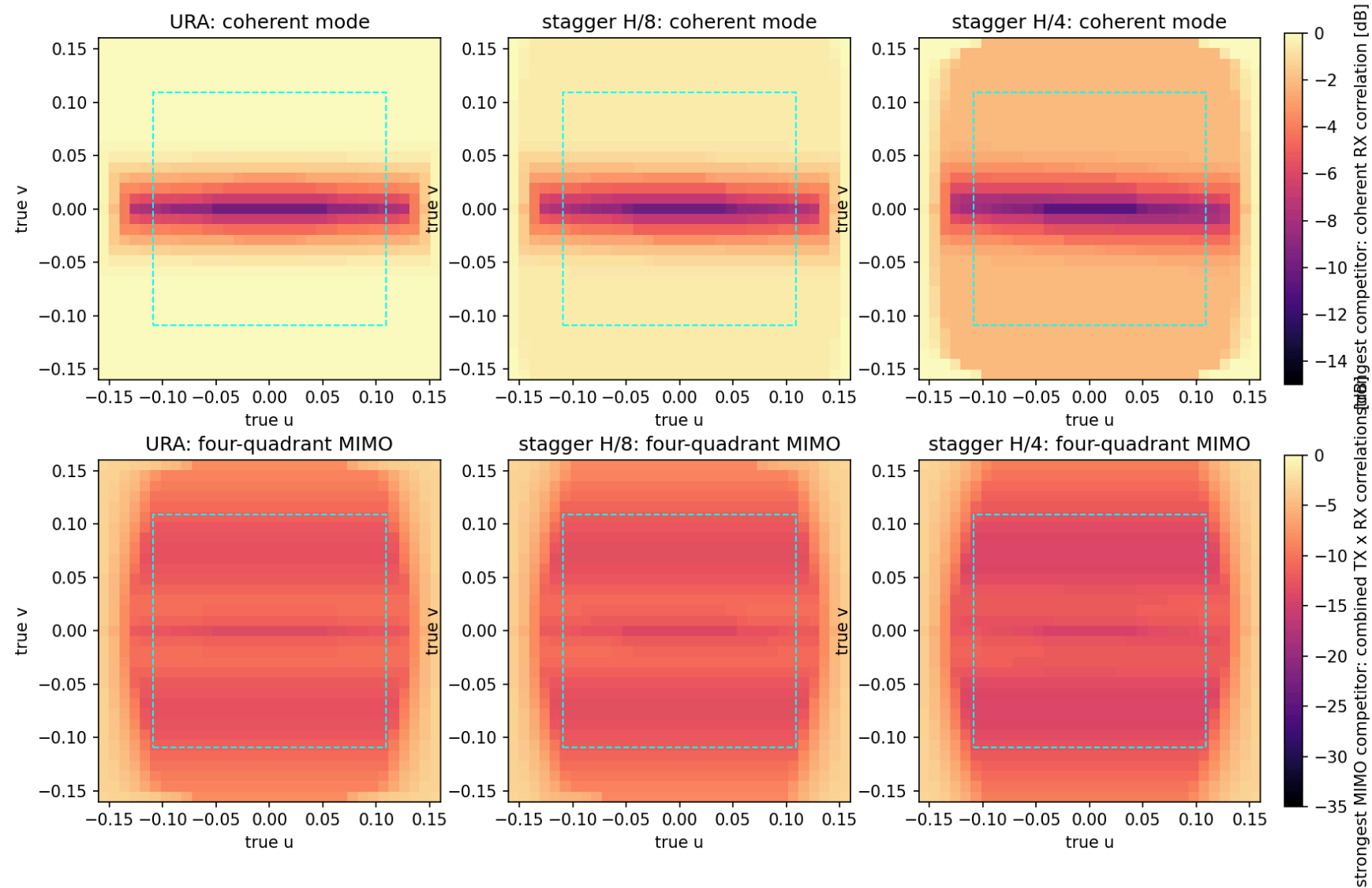
Whole-beam check: strongest competitor per true direction

Within the TX 3 dB region, competitors outside the own lobe needing at most +10 dBsm:

Layout	Worst coherent competitor	Directions with one ≥ -1 dB	MIMO correlation at that competitor
URA	0.0 dB	57%	-10.3 dB
H/8	-0.5 dB	51%	-10.6 dB
H/4	-2.2 dB	0%	-11.4 dB

- H/4 leaves nothing near-exact anywhere in the beam in coherent mode
- The last column is the MIMO correlation *of the strongest coherent competitor*, not the worst MIMO competitor: another direction can be more competitive in MIMO (review example: URA at (0, 0.10), about -12 dB at the strongest MIMO competitor versus -27 dB at the selected one). No global MIMO guarantee is claimed; the worst-MIMO map is being added

Strongest alias competitor for every in-beam true direction
 competitors outside the own old-URA cell needing at most +10 dBsm, searched separately per mode; cyan: TX 3 dB region



Layout options (from the requirements draft)

Option	Subarray height	Stagger	RX height	Vertical edge	1 - rho	RX gain
A. URA	18.81 mm (8 rows)	0	37.6 mm	$\pm 5.9^\circ$	0.00	0 dB
B. One pitch (our sketch)	18.81 mm (8 rows)	2.35 mm	40.0 mm	$\pm 5.9^\circ$	0.15	0 dB
C. Two pitches	18.81 mm (8 rows)	4.70 mm	42.3 mm	$\pm 5.9^\circ$	0.50	0 dB
D. Two pitches, 7 rows	16.46 mm (7 rows)	4.70 mm	37.6 mm	$\pm 6.8^\circ$	0.61	-0.58 dB

- D keeps the original 16-row grid, pitch and total height: each column populates 14 of 16 rows, offset two rows between adjacent columns; costs about 3% range
- D is a promising extrapolation: not yet run through the event simulation, and its changed height also moves its aliases relative to the TX pattern
- B gives too little separation to be worth its complexity

Recommendation

Two-pitch alternating column stagger: option C, with D as a variant to validate. URA as fallback.

Why:

1. The vertical alias is the operationally important one (in-beam, long range)
2. For a URA, coherent frames carry nothing about it; nearly the whole beam relies on MIMO
3. MIMO's discrimination is a property of the TX defocus, calibration and waveform, not of geometry
4. A two-pitch stagger gives every 20 Hz frame half the separation of an unambiguous measurement, at no gain cost
5. It leaves no near-exact competitor anywhere in the beam
6. Its evidence is RX geometry only, independent of the TX realization

What stays: MIMO for the horizontal family (short range, high SNR) and as fallback, **on demand**, not interlaced. The -3 dB horizontal-edge discrimination from the TX defocus is the binding TX requirement.

Treat ambiguity-resolution latency, calibration tolerance and option D's detailed performance as **validation items, not settled specifications.**

Risks and costs

Stagger:

- Non-separable steering: cost per formed beam unchanged at 8 channels; beam-bank size, search and likelihood updates not yet budgeted (likely manageable). Design bookkeeping: lobes on a sheared lattice, 2-D validation, per-lobe likelihoods in the tracker
- Relative RX channel phase: margin 3.0 dB nominal, 1.9 dB at $+15^\circ$, zero at $+45^\circ$ systematic column-group error. $\pm 15^\circ$ is a proposed calibration target, not demonstrated production robustness
- Option C adds 4.7 mm height; option D costs 0.6 dB

MIMO (unchanged by the layout, but now carrying less of the load):

- Supplier must realize the quadrant squint; intra- and inter-quadrant calibration; waveform orthogonality and its Doppler cost; pattern-dictionary processing

Not yet shown for any layout: slow RCS fluctuation across frames, multi-target association, decision rule with abstention

Fallbacks if MIMO underperforms (ideas, not analysed): quadrant-level TX sequential lobing with the coherent

Antenna specification: what the supplier must be told

- The four individual TX quadrant patterns are a **requirement**, not a by-product of the sum beam
- Reference excitation to aim for: the supplied amplitude **and** quadratic phase (`antenna_arr_77_TX_rev_A.mat`); any other beam-widening mechanism must be agreed against the metrics below
- Deliverable: complex embedded patterns for each of the 8 TX ports and 8 RX channels, common phase reference, $|u|, |v| \leq 0.6$ at 0.005, centre and band-edge frequencies; simulated at review, measured on the prototype
- Acceptance metrics we compute (four-quadrant correlation):

Direction pair	Requirement	Nominal
Opposite vertical edges (0, ± 0.1035)	≤ -10 dB	-29 dB
Opposite horizontal edges (± 0.207 , 0)	≤ -2.5 dB	-3.1 dB
Boresight vs first vertical replica	≤ -6 dB	-8.2 dB

- Stability, not tolerance: static offsets are calibrated; ask for expected inter-quadrant and inter-channel phase variation over temperature

Asks for this meeting

1. Agree the direction: two-pitch stagger (C or D), URA fallback
2. Agree to send the supplier the requirements draft and a shortlist (A, C, D) for **early feedback on obvious issues**, not a full analysis
3. Check the measured per-port patterns of the existing four-quadrant prototype against the quadrant metrics as a first realizability test
4. Confirm the TX excitation is specified deliberately: quadratic defocus or agreed equivalent, quadrant patterns as deliverables
5. Requirements still needed: which alias family drives (long-range vertical vs short-range horizontal), acceptable wrong-cell publication probability, and publishing tracks with explicit ambiguity to higher layers

Backup: reproduction

```
source venv/bin/activate
python studies/2026-09-02_lannik-psi/rx_layout_experiment.py
python studies/2026-09-02_lannik-psi/rx_resolution_experiment.py
python studies/2026-09-02_lannik-psi/rx_stagger_amount_experiment.py    # about 3 min
python -m pytest studies/2026-09-02_lannik-psi -q
```

Figures under `generated/experimental/` and `generated/experimental/on_demand/`.

Model limits: ideal orthogonal waveforms, exact nominal patterns, fixed 1 m² target known to exist in a gate, independent frame noise, no motion, no tracker, no measured latency; energies are illumination equivalents, not wall-clock time.